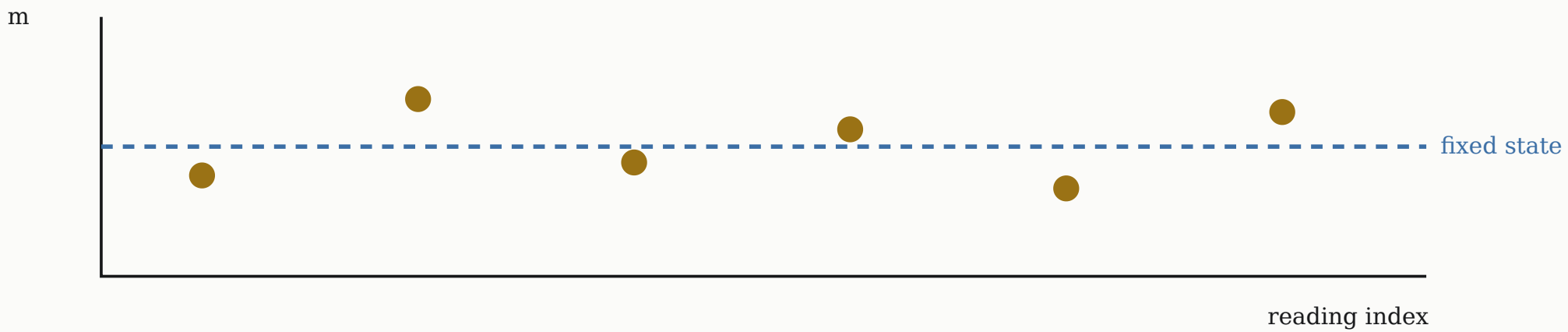
MOTIVATION

From Noisy Readings to Reliable Estimates

Which position should a noisy sensor report?



> One stationary object. Several different readings.

`state` · `measurements` · `inferred answer`

Report an estimate and the uncertainty around it.

Mathematics for Robotics · ROB-GY 6013 · NYU Tandon · Jing Zhang · Fall 2026 · November 4

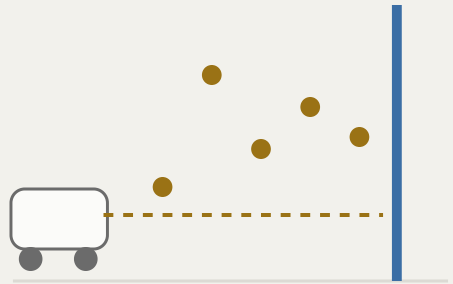
Probability model; parameter estimation — [1] DL §3.1 · [1] DL §5.5.1 · Course example

ROADMAP

From Readings to Estimates

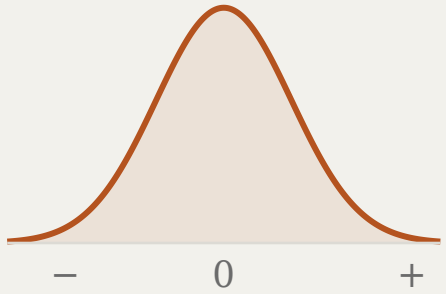
How does the lecture fit together?

Readings



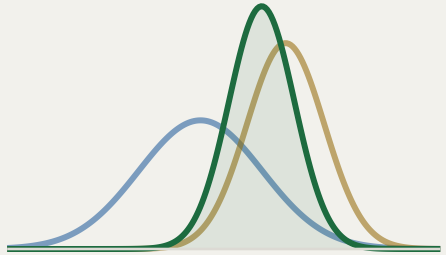
Same wall, noisy readouts

Error



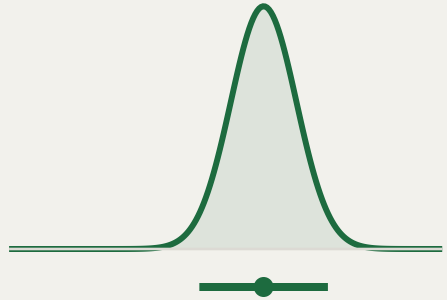
Model the error

Update



Combine information

Estimate



Report a value and a range

> What should we report? How sure are we?

Probability distribution; uncertainty — [1] DL §3.1 · Course concept map

DEFINITION

Random variables

A random variable assigns a real number to each outcome in the sample space.

$$x : \Omega \to \mathbb{R}, \qquad x(\omega) \in \{-1, 0, 2\}$$

- Ω sample space: all possible outcomes
- ω one outcome
- x random error (cm)

Outcome A $\to -1$ cm

Outcome B $\to 0$ cm

Outcome C $\to 2$ cm

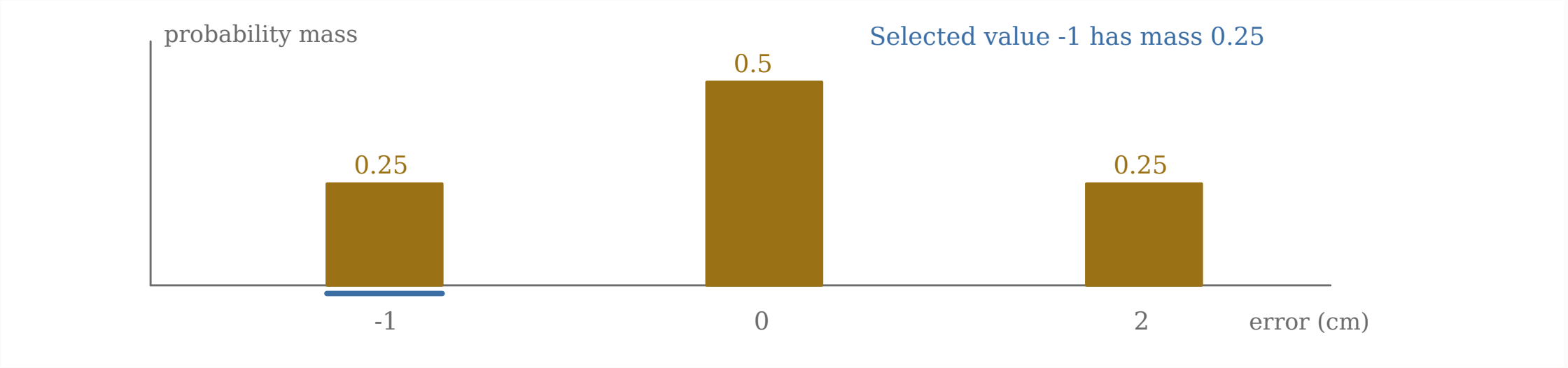
> A constant random variable assigns every outcome the same number.

Random variable; realization — [10] MC Probability §5.2

DEFINITION

Probability Mass Function (PMF)

A PMF assigns a probability to each value of a discrete random variable.



$$p(a) = P(x = a), \qquad p(a) \geq 0, \qquad \sum_{a} p(a) = 1$$

- x discrete error: finitely or countably many values (cm)
- a possible value (cm)
- $p(a)$ probability of $x = a$
- P event probability; an event is a set of outcomes

Probability mass function (PMF) — [1] DL §3.3.1 · [7] Data140 §1.1

EXAMPLE

Probability Mass Function (PMF)

The mean is the probability-weighted average error.

Error values −1, 0, 2 cm; masses ¼, ½, ¼. Find nonnegative-error probability and mean.

$$P(x \geq 0) = \tfrac{1}{2} + \tfrac{1}{4} = \tfrac{3}{4}, \qquad \mu = \sum_a a\, p(a) = (-1)\tfrac{1}{4} + 0\tfrac{1}{2} + 2\tfrac{1}{4} = \tfrac{1}{4}$$

- P event probability
- x error (cm)
- μ mean error; sensor bias (cm)

Discrete probability; expectation — [1] DL §3.3.1 · [7] Data140 §§8.1,8.3 · Course example

EXERCISE

Probability Mass Function (PMF)

Which values belong to the event?

Errors $-1, 0, 2$ cm have masses $1/2, 1/4, 1/4$. What is $P(x \geq 0)$?

A $1/4$: count only positive errors.

B $1/2$: include zero and positive errors.

C $2/3$: two of three possible values.

Event probability; PMF — [1] DL §3.3.1 · Course question

DEFINITION

Probability Density Function (PDF)

A PDF gives interval probabilities as areas under a nonnegative density curve.

$$p(t) \geq 0, \quad \int_{-\infty}^{\infty} p(t)\,dt = 1$$

$$P(a \leq x \leq b) = \int_{a}^{b} p(t)\,dt$$

- x random error (cm)
- a, b interval endpoints, $a \leq b$
- t integration variable
- $p(t)$ density (cm^{-1})

$$P(x = c) = \int_{c}^{c} p(t)\,dt = 0$$

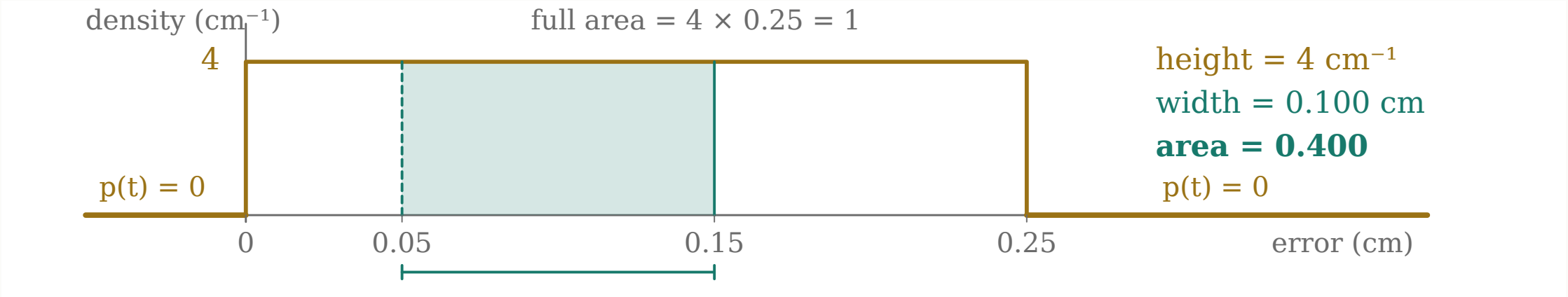
- c fixed error value

Probability density function (PDF) — [1] DL §3.3.2

EXAMPLE

Probability Density Function (PDF)

A uniform density is constant on its interval and zero outside it.



UNIFORM ERROR

$$p(t) = \begin{cases} 4 & 0 \le t \le .25 \\ 0 & \text{otherwise} \end{cases}, \quad P(.05 \le x \le .15) = 4(.15 - .05) = .4$$

t, x error (cm) p density (cm^{-1}) P interval probability

Uniform density; normalization — [1] DL §3.3.2 · Course example

EXERCISE

Probability Density Function (PDF)

Density is height; probability is area

x is uniform on $[0, .5]$ cm. Which statement is valid?

A. Density 2 cm^{-1}; $P(.1 \leq x \leq .3) = .4$.

B. Density 2 means event probability 2.

C. Probability is $.2$, the interval width alone.

Density; event probability — [1] DL §3.3.2 · Course question

DEFINITION

Moments

Expectation is the probability-weighted average of a function’s values.

$$\mathbb{E}[g(x)] = \sum_{a} g(a)p(a)$$

- x discrete variable
- a possible value
- g function; assume $\mathbb{E}[|g(x)|] < \infty$
- p PMF; moments average powers: mean $\mathbb{E}[x]$, second moment $\mathbb{E}[x^2]$

$$\mathbb{E}[g(x)] = \int_{-\infty}^{\infty} g(t)p(t)\,dt$$

- x continuous variable
- t integration variable
- p density
- g same function

Expectation of a function — [1] DL §3.8 · [7] Data140 §§8.1,8.3 · Course derivation

EXERCISE

Moments

Variance is the average squared deviation from the mean.

$\mathrm{Var}(x) = \mathbb{E}[(x-\mu)^2]$. Given $\mu = 1/4$ cm and $\mathbb{E}[x^2] = 5/4$ cm^2, choose the variance.

A $5/4$: the second moment is always variance.

B 1: subtract the mean, not its square.

C $19/16$: subtract the squared mean.

Centering; variance — [1] DL §3.8 · Course question

DEFINITION

Moments

The mean vector lists the component expectations.

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \qquad \mu = \mathbb{E}[x] = \begin{bmatrix} \mathbb{E}[x_1] \\ \mathbb{E}[x_2] \end{bmatrix}$$

- x column of random variables on one outcome space
- x_i component i
- μ mean in R^2; finite component means

> Average components; keep the column shape.

Sample index s: $x^{(s)}$. Component index i: x_i.

Random vector; mean vector — [7] Data140 §23.1

DEFINITION

Moments

Covariance averages products of deviations from the component means.

$$\Sigma = \mathbb{E}[(x-\mu)(x-\mu)^T], \qquad \Sigma_{ij} = \mathbb{E}[(x_i-\mu_i)(x_j-\mu_j)]$$

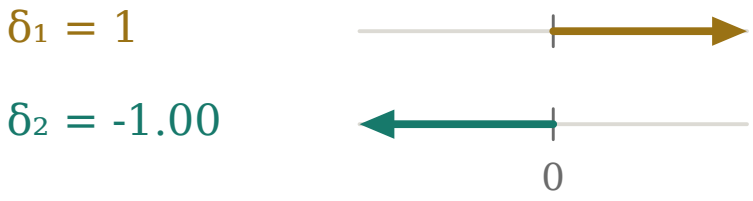
x, μ random column and mean in $\mathbb{R}^d$; assume $\mathbb{E}[x_i^2] < \infty$

Σ population covariance, d × d

i, j coordinate indices

One centered outcome: δ = x − μ

dimensionless illustration



One outer product $\delta\delta^T$

$\delta_1^2 = 1$	$\delta_1\delta_2 = -1.00$
$\delta_2\delta_1 = -1.00$	$\delta_2^2 = 1.00$

Σ averages these matrices over the population.

> In Σ: diagonal = component variance; off-diagonal = mean paired product.

Covariance matrix — [1] DL §3.8 · [7] Data140 §23.1 · Course derivation

DERIVATION

Moments

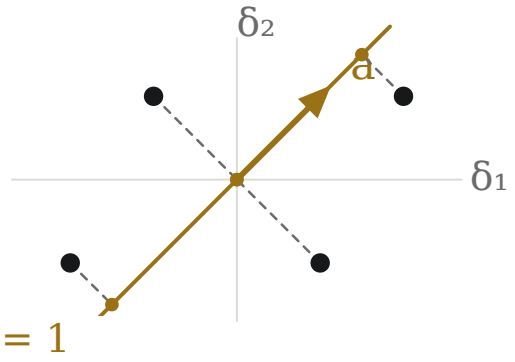
Positive semidefinite means symmetric, with $a^T \Sigma a \geq 0$ for every a.

$$a^T \Sigma a = \mathbb{E}[(a^T(x-\mu))^2] \geq 0$$

- a fixed direction column in R^d
- x, μ random column and population mean
- Σ population covariance (PSD)

$$C = \frac{X_c^T X_c}{N-1} \quad (N > 1)$$

- C sample covariance estimate
- X_c N sample rows, each minus the sample mean
- N number of samples



Centered population: each point has probability ¼

δ = x − μ	y = aᵀδ	y² ≥ 0
(2, 1)	2.12	4.50
(-2, -1)	-2.12	4.50
(1, -1)	0.00	0.00
(-1, 1)	0.00	0.00

Var(y) = average of y² = 2.250 ≥ 0

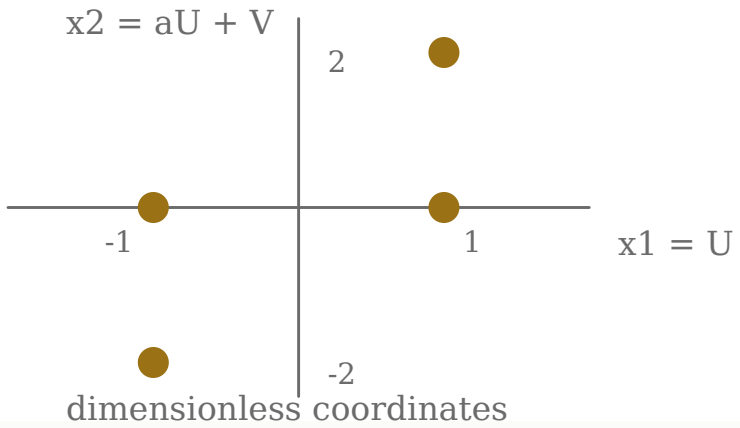
dimensionless model

Positive semidefinite covariance — [1] DL §3.8 · [1] DL §2.7 · [7] Data140 §23.1 · Course derivation

EXAMPLE

Moments

Independence means knowing one sign does not change the other's probabilities.



a = 1.00

four equally likely outcomes

a: 0 to 1 adds the shared sign

Independent fair signs: $P(U=u,V=v)=P(U=u)P(V=v)$ for all $u,v=\pm 1$. Let $x=(U,U+V)^T$.

$$\mu=\begin{bmatrix}0\\0\end{bmatrix},\qquad \Sigma=\begin{bmatrix}1&1\\1&2\end{bmatrix}$$

U,V fair signs: ±1, each with mass ½

x,μ random column and mean

Σ covariance; a nonzero off-diagonal entry means correlation

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · Course example

EXERCISE

Moments

Independent sources, dependent components

U, V are independent fair signs. Let $x = (U, 2U + V)^T$. What is $\mathrm{Cov}(x_1, x_2)$?

A. 2: both components share U.

B. 0: independent sources imply independent outputs.

C. 1: doubling a coefficient has no effect.

Independence; covariance — [1] DL §3.7 · [1] DL §3.8 · [1] DL §2.7 · Course question

COUNTEREXAMPLE

Moments

Uncorrelated means zero covariance; independence is a stronger condition.

x is uniform on $\{-1,0,1\}$; $y=x^2$. Is y independent of x?

$$\mathrm{Cov}(x,y)=0, \qquad P(y=0 \mid x=0)=1 \neq \tfrac{1}{3}=P(y=0)$$

x, y	random scalars
$P(y=0 \mid x=0)$	probability given $x=0$: $P(y=0,x=0)/P(x=0)$
Cov	population covariance

Uncorrelatedness; independence — [1] DL §3.8 · [2] UBC Lec.18 p.8 · Course example

DEFINITION

Gaussian geometry

A vector is Gaussian if every linear combination is normal (constant at zero variance).

$$p(x) = (2\pi)^{-d/2}|\Sigma|^{-1/2}\exp(-d_M^2/2)$$

- x, μ — point/mean in R^d
- d — dimension
- $\Sigma \succ 0$ — $a^T\Sigma a > 0$ for every $a \neq 0$
- $|\Sigma|$ — determinant
- p — density
- d_M^2 — squared covariance-scaled (Mahalanobis) distance

$$d_M^2 = (x-\mu)^T\Sigma^{-1}(x-\mu)$$

- Σ^{-1} — inverse covariance
- $x - \mu$ — deviation

Multivariate Gaussian distribution — [1] DL §3.9.3 · [7] Data140 §23.2

DERIVATION

Gaussian geometry

Whitening centers a vector and gives it identity covariance.

Derivation

Derive the mean and covariance

$\mu = \mathbb{E}[x]$; $\Sigma = \mathrm{Cov}(x) = Q\Lambda Q^T \succ 0$; $Q^TQ = I$; Λ positive diagonal. Fixed whitening map $W = \Lambda^{-1/2}Q^T$; $z = W(x - \mu)$; I: identity.

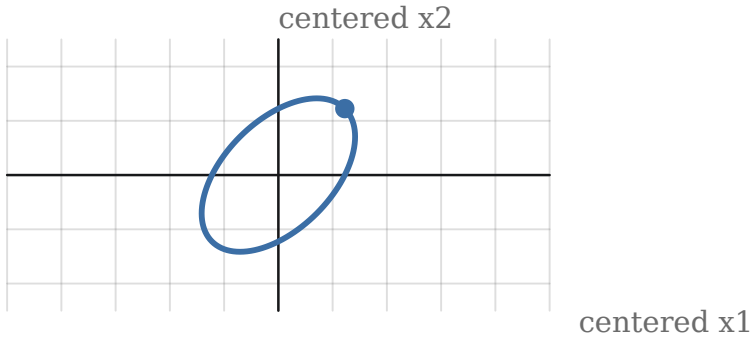
1. $\mathbb{E}[z] = W(\mathbb{E}[x] - \mu) = 0$ by linearity.
2. $\mathrm{Cov}(z) = \mathbb{E}[zz^T] = W\,\mathbb{E}[(x-\mu)(x-\mu)^T]\,W^T = W\Sigma W^T$.
3. $= \Lambda^{-1/2}Q^T(Q\Lambda Q^T)Q\Lambda^{-1/2} = \Lambda^{-1/2}\Lambda\Lambda^{-1/2} = I$.
4. Fixed matrices pass through expectation; $Q^TQ = I$ cancels the basis changes.

Whitening; eigendecomposition — [6] Stanford EE278 pp.3–10–12 · [2] UBC Lec.18 p.5 · Course derivation

DERIVATION

Gaussian geometry

A density contour joins points with the same density.



unit Mahalanobis contour

eigenvalues 3 and 1

$$x = \mu + Q\Lambda^{1/2} z, \qquad \|z\| = c$$

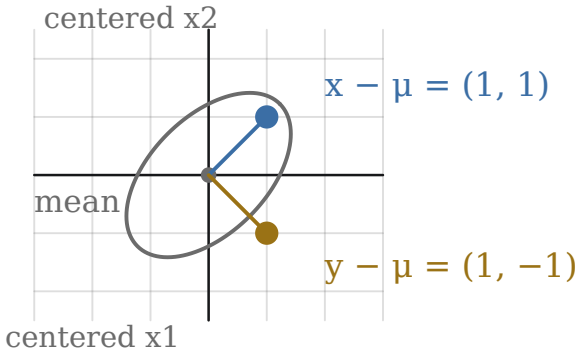
$c > 0$: standardized radius; q_i: unit eigenvectors (columns of Q). Semiaxis: center-to-boundary length along q_i, equal to $c\sqrt{\lambda_i}$. Plot of $x - \mu$: $c = 1$, $\lambda = (3, 1)$, $q_1 = (1, 1)^T/\sqrt{2}$, $q_2 = (1, -1)^T/\sqrt{2}$.

Gaussian density contour — [2] UBC Lec.18 p.5 · Course derivation

EXAMPLE

Gaussian geometry

Equal Euclidean lengths can have different Gaussian densities.



gray ellipse: $d_m = 1$

both arrows: Euclidean length $\sqrt{2}$

$$\Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad \mu = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Σ covariance; μ mean; compare $x = (2, 0)^T$ and $y = (2, -2)^T$.

$$d_M^2(x) = \tfrac{2}{3}, \qquad d_M^2(y) = 2$$

d_M covariance-scaled distance.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course example

EXERCISE

Gaussian geometry

Which point has the higher Gaussian density?

$\mu = 0$, $\Sigma = \mathrm{diag}(4, 1)$; $u = (2, 0)^T$, $v = (0, 2)^T$. Compare Gaussian densities.

A $p(u) = p(v)$: their Euclidean lengths agree.

B $p(u) > p(v)$: u is one standard deviation away.

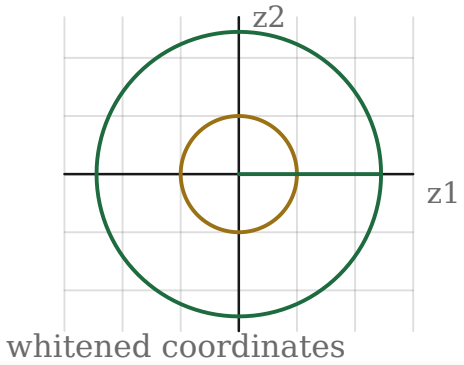
C $p(u) < p(v)$: higher variance always lowers density.

Mahalanobis distance — [2] UBC Lec.18 p.5 · Course question

DERIVATION

Gaussian geometry

Choose a region by the probability it encloses.



c = 1: mass 39.3%

c = 2.448: mass 95%

original semiaxes: 4.24 and 2.45

circle radius = Mahalanobis distance

$$P(d_M^2 \le c^2) = 1 - e^{-c^2/2} \quad (d = 2)$$

Here the region contains all points with $d_M \le c$; its probability is the integral of density inside. $d = 2$; nonsingular Gaussian. Example: $\lambda_1 = 3, \lambda_2 = 1$.

$$c_{.95} = \sqrt{-2 \log .05} \approx 2.44775$$

$c_{.95}$ radius enclosing 95%.

Chi-square distribution; Gaussian probability region — [3] NIST §1.3.6.6.6 · [3] NIST χ^2 table · Course derivation

DEFINITION

Bayes

Conditional probability is the chance of an event given another.

joint events	positive +	negative −	row total
event H	H and +	H and −	P(H)
no event	not H and +	not H and −	P(not H)
	total P(+)	total P(−)	

$$P(H \mid +) = \frac{P(H \cap +)}{P(+)}$$

$$P(+ \mid H) = \frac{P(H \cap +)}{P(H)}$$

H: event; $+$: positive reading; $\cap$: both occur. Sensitivity $P(+ \mid H)$: positive rate among event cases. Posterior $P(H \mid +)$: event probability after a positive. Denominators must be positive.

Conditional probability — [1] DL §3.4–3.5 · [9] Wisconsin Math225 Lec.5 · Course example

DERIVATION

Bayes

Bayes’ rule updates a prior probability using an observation.

$$P(H \mid +) = \frac{P(+ \mid H)P(H)}{P(+)}$$

$P(H)$ prior: event chance before this reading. $P(+ \mid H)$ likelihood: positive-reading chance if H holds. $P(+) > 0$ evidence: overall positive-reading chance. $P(H \mid +)$ posterior: event chance after this reading.

Bayes’ rule — [1] DL §3.11 · [8] CMU Bayes notes p.1 · Course derivation

DERIVATION

Bayes

Total probability adds contributions from disjoint cases.

$$P(+) = P(+ \mid H)P(H) + P(+ \mid \neg H)P(\neg H)$$

A partition has disjoint cases covering all trials: here H (event) and $\neg H$ (no event). Their joint events with $+$ cover all positive readings. $P(+)$: evidence.

Law of total probability; evidence — [1] DL §3.4–3.5 · [1] DL §3.11 · Course derivation

EXAMPLE

Bayes

Why can most positive alarms be false?

joint events	positive +	negative −
event H	90	10
no event	990	8910

$$P(H) = .01, \quad P(+ \mid H) = .9, \quad P(+ \mid \neg H) = .1$$

$P(H) = .01$ prevalence: event probability before this reading. Expected counts in 10,000 model trials: 100 event cases and 9,900 non-event cases.

$$P(H \mid +) = \frac{90}{90 + 990} = \frac{1}{12}$$

H event; $+$ positive reading.

Prior probability; posterior probability — [1] DL §3.11 · [9] Wisconsin Math225 Lec.5 · Course example

EXERCISE

Bayes

How does prevalence change the chance an alarm is true?

$P(H) = .1$, $P(+ \mid H) = .9$, $P(+ \mid \neg H) = .1$. What is $P(H \mid +)$?

A .9: the chance of a true event equals sensor sensitivity.

B .1: an observation cannot change the prior.

C .5: true and false positives contribute equally.

Prior probability; posterior probability — [1] DL §3.11 · Course question

RECAP

Break

Pause: probability, density, and conditioning

$$P(X \in A), \qquad p(x), \qquad P(H \mid +)$$

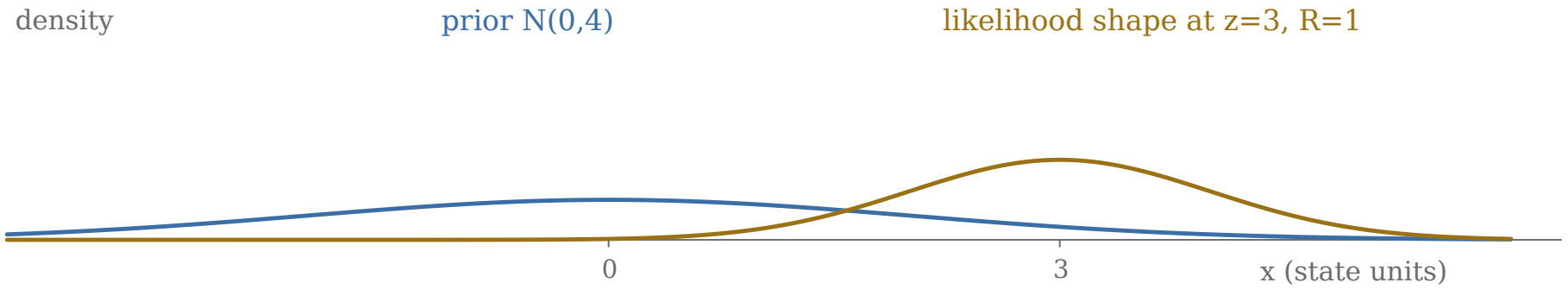
X: random quantity (written x earlier); lowercase x: evaluation value here. A: set of values; $\{X \in A\}$: event; p: density. H: event of interest; $+$: positive reading. Conditioning changes the reference population.

Event probability; density; conditional probability — [1] DL §3.3.2 · [1] DL §3.4–3.5 · Course recap

MODEL

Gaussian conditioning

A posterior density describes uncertainty after a reading.



$$x \sim \mathcal{N}(\mu_0, P_0), \quad z = x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, R)$$

x: state; z: reading. $P_0, R > 0$: prior/noise variances; ε independent of x. Prior $p(x)$: density before observation; likelihood $p(z \mid x)$: reading density for candidate x; posterior $p(x \mid z)$: updated state density.

Gaussian prior; Gaussian likelihood — [2] UBC Lec.18 p.7 · [8] CMU Bayes notes p.1

DERIVATION

Gaussian conditioning

Precision is inverse variance.

$$J(x) = \frac{(x-\mu_0)^2}{2P_0} + \frac{(z-x)^2}{2R}, \quad P_1^{-1} = P_0^{-1} + R^{-1}, \quad \mu_1 = P_1(\mu_0/P_0 + z/R)$$

J: negative log posterior, up to a constant. Independent Gaussian prior/noise; $P_0, R > 0$. x: state; z: reading. μ_0, P_0: prior mean/variance; R: noise variance; μ_1, P_1: posterior mean/variance.

$$a = P_0^{-1} + R^{-1}, \qquad b = \mu_0/P_0 + z/R$$
$$J(x) = \tfrac{1}{2}(ax^2 - 2bx) + C = \tfrac{a}{2}(x - b/a)^2 + C\prime$$

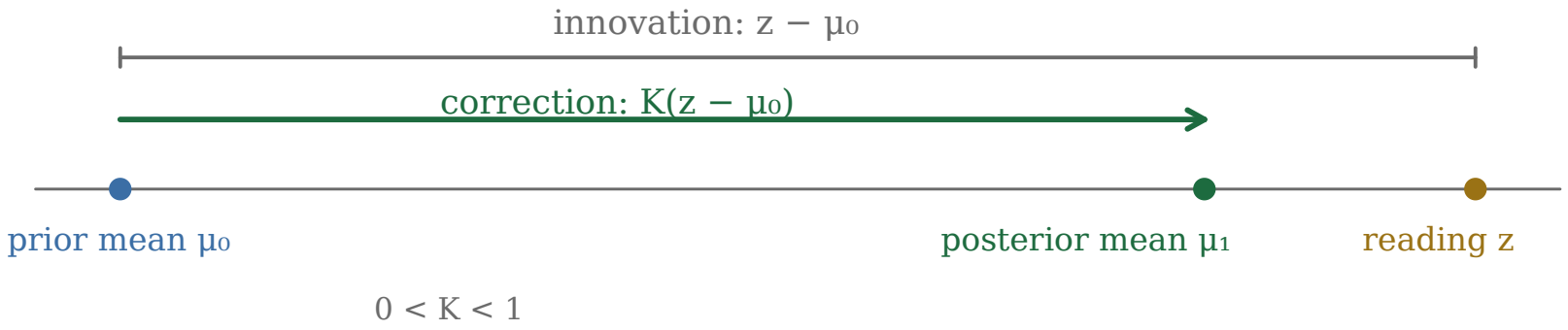
a, b scalar coefficients; $C, C\prime$: constants independent of x.

Gaussian posterior; precision — [2] UBC Lec.18 p.7 · [1] DL §3.9.3 · Course derivation

INTERPRETATION

Gaussian conditioning

The gain scales the innovation to update the mean.



$$\mu_1 = \frac{R\mu_0 + P_0 z}{P_0 + R} = \mu_0 + K(z - \mu_0), \quad K = \frac{P_0}{P_0 + R}, \quad P_1 = (1 - K)P_0$$

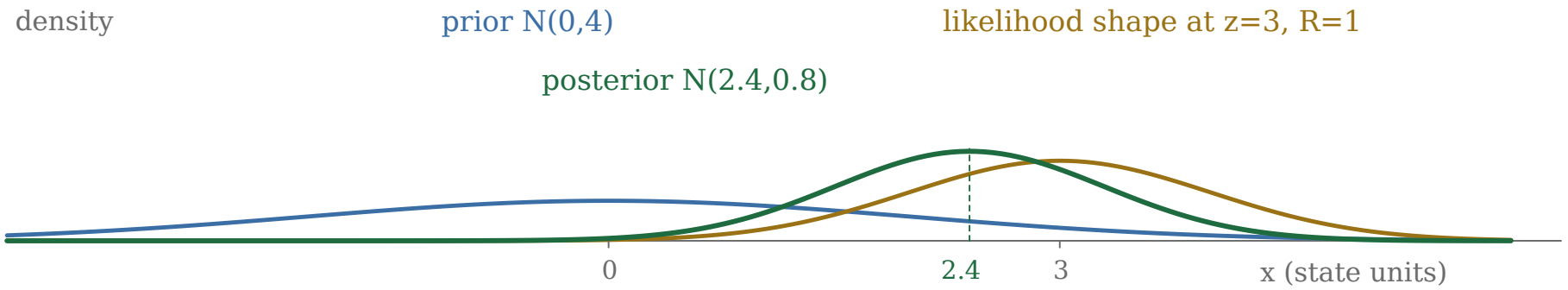
μ_0, P_0: prior mean/variance; z: reading; R: noise variance. μ_1, P_1: posterior mean/variance. Innovation $z - \mu_0$: observed minus predicted reading. Gain K: fraction of that discrepancy used, with $0 < K < 1$.

Kalman gain; measurement innovation — [5] Welch–Bishop §1 · [2] UBC Lec.18 p.7 · Course derivation

EXAMPLE

Gaussian conditioning

Fuse a prior and a precise reading



$$\mu_0 = 0, \quad P_0 = 4, \quad z = 3, \quad R = 1$$

μ_0: prior mean; P_0: prior variance; z: reading; R: independent noise variance.

$$K = \tfrac{4}{5}, \qquad \mu_1 = 2.4, \qquad P_1 = .8$$

K gain; μ_1, P_1 posterior mean, variance.

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course example

EXERCISE

Gaussian conditioning

What does a noisier reading change?

Prior $\mathcal{N}(0, 4)$; $z = 3$ with independent Gaussian noise, variance R: $1 \to 4$. Compare posterior mean and variance.

A. Mean moves toward 0; posterior variance increases.

B. Mean moves toward 3; posterior variance decreases.

C. Only the mean changes; variance ignores sensor noise.

Gaussian conditioning — [2] UBC Lec.18 p.7 · Course question

RESULT

Gaussian conditioning

A conditional mean is an expectation given an observation.

$$\mu_{a|b} = \mu_a + \Sigma_{ab}\Sigma_{bb}^{-1}(b - \mu_b)$$

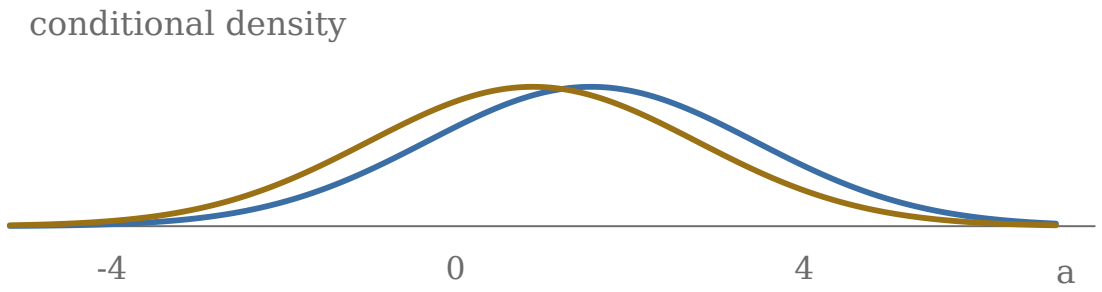
Joint Gaussian: normal linear combinations (or constants). a: inferred p-vector; b: observed q-vector; μ_a, μ_b: marginal means. Σ_{bb}: invertible covariance of b. Cross-covariance averages centered products: $\Sigma_{ab} = \mathbb{E}[(a - \mu_a)(b - \mu_b)^T]$.

Conditional Gaussian mean — [2] UBC Lec.18 p.7 · [7] Data140 §23.2

RESULT

Gaussian conditioning

Conditional covariance measures spread after an observation.



b = 5: mean 5/3

b = 2: mean 1

both variances 32/9

$$\Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$$

a: inferred block; b: observed block. Σ_{aa}: covariance of a; $\Sigma_{a|b}$: covariance after observing b. Joint Gaussian, $\Sigma_{bb} \succ 0$, $\Sigma_{ba} = \Sigma_{ab}^T$. Plot: $\mu = (1, 2)^T$, $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}$.

Conditional Gaussian covariance — [2] UBC Lec.18 p.7 · [2] UBC Lec.18 p.8

EXERCISE

Gaussian conditioning

Does the conditional variance change?

Gaussian $(a, b)^T$: $\mu = (1, 2)^T$, $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 9 \end{bmatrix}$. Observe $b = -1$ instead of 5.

A. Both conditional mean and variance decrease.

B. Mean decreases; conditional variance stays fixed.

C. Neither changes because the joint model is fixed.

Conditional Gaussian distribution — [2] UBC Lec.18 p.7 · Course question

DEFINITION

Estimation

Maximum likelihood estimation (MLE) maximizes data likelihood.

$$b = A\theta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$$

$$L(\theta; b) = p(b \mid \theta),\ \hat{\theta}_{MLE} \in \arg\max_{\theta} L(\theta; b)$$

b: data N-vector; A: known N×p; θ: parameter p-vector; $\sigma^2 > 0$: known variance. L: density of fixed b as θ varies; argmax: maximizers

Gaussian linear regression; maximum likelihood — [1] DL §5.5

DERIVATION

Estimation

Gaussian products become a squared loss

$$-\log p(b \mid \theta) = \frac{N}{2}\log(2\pi\sigma^2) + \frac{\|b - A\theta\|^2}{2\sigma^2}$$

A known; b observed; θ unknown; independent errors, each $\mathcal{N}(0, \sigma^2)$, $\sigma^2 > 0$.

Negative log-likelihood; least squares — [1] DL §5.5.1 · Course derivation

DERIVATION

Estimation

Normal equations set the least-squares gradient to zero.

$$A^T A\hat{\theta} = A^T b, \qquad A^T(b - A\hat{\theta}) = 0$$

$A \in \mathbb{R}^{N\times p}$: known design; $b \in \mathbb{R}^N$: data; $\theta \in \mathbb{R}^p$: parameters. Independent equal-variance Gaussian errors make $\hat{\theta}$ the MLE. Full column rank means independent columns and gives uniqueness; the residual is orthogonal to every column.

Normal equations; least squares — [1] DL §5.1.4 · Course derivation

EXAMPLE

Estimation

Fit three readings with two coefficients

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$\theta = (\text{intercept}, \text{slope})^T$; $r = b - A\hat{\theta}$.

$$\hat{\theta} = \begin{pmatrix} 7/6 \\ 1/2 \end{pmatrix}, \quad r = \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix}$$

$\hat{\theta}$ estimate; r observed minus fitted.

Least-squares estimate — [1] DL §5.1.4 · Course example

DEFINITION

Estimation

Maximum a posteriori estimation (MAP) maximizes the posterior density.

$$J(\theta) = \tfrac{1}{2}\|b - A\theta\|^2 + \tfrac{\lambda^2}{2}\|\theta - \mu_0\|^2, \quad \lambda^2 = \sigma^2/\tau^2$$

Minimize J for MAP under p.38's Gaussian noise model. Prior $\theta \sim \mathcal{N}(\mu_0, \tau^2 I)$, $\tau^2 > 0$: equal variance in every unit direction. $\sigma^2 > 0$: known variance; λ^2: quadratic penalty weight.

Maximum a posteriori (MAP); quadratic regularization — [1] DL §5.6.1 · Course derivation

EXAMPLE

Estimation

The prior changes the normal equation

$$(A^T A + \lambda^2 I)\hat{\theta} = A^T b + \lambda^2 \mu_0$$

Example: $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$, $b = (1, 2, 2)^T$, $\mu_0 = 0$, $\lambda^2 = 1$.

$$\hat{\theta}_{MAP} = \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix}, \quad \|b - A\hat{\theta}\|^2 = \tfrac{2}{5}$$

$\hat{\theta}_{MAP} = (\text{intercept}, \text{slope})^T$: posterior mode; $\|b - A\hat{\theta}\|^2$: squared residual norm.

MAP normal equations; ridge regularization — [1] DL §5.6.1 · [1] DL §5.1.4 · Course derivation

EXERCISE

Estimation

What makes the MAP estimate unique?

$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$, $b = (1, 2)^T$; $J = \frac{1}{2}\|b - A\theta\|^2 + \frac{1}{2}\|\theta\|^2$.

A. The prior makes the two data columns independent.

B. Gaussian noise makes every data fit unique.

C. The prior adds curvature along the data null space.

MAP estimate; positive definiteness — [1] DL §5.6.1 · [1] DL §2.7 · [1] DL §4.3 · Course question

EXERCISE

Sensor Calibration

Calibration: choose the likelihood

$b = (1, 2, 3)^T$: readings minus a known reference; $b_i = \theta + \varepsilon_i$, $\varepsilon_i \overset{iid}{\sim} \mathcal{N}(0, 1)$. Which data objective fits this model?

A $\frac{1}{2}\sum_i (b_i - \theta)^2$

B $\sum_i |b_i - \theta|$

C $\frac{1}{2}\sum_i (b_i - \theta)^2 + \frac{1}{2}\theta^2$

Gaussian likelihood; maximum likelihood — [1] DL §5.5.1 · Course example

EXERCISE

Sensor Calibration

Calibration: combine data and prior

Readings $(1, 2, 3)$; independent normal noise, variance 1; prior $\theta \sim \mathcal{N}(0, 1)$. Find posterior mean and precision (inverse variance).

A. Center stays at 2; only uncertainty changes.

B. Center moves toward 0; precision becomes 4.

C. Center moves toward 0; variances add to 4.

Gaussian posterior; MAP estimate — [2] UBC Lec.18 p.7 · [1] DL §5.6.1 · Course example

EXERCISE

Sensor Calibration

Calibration: confidence coverage

Three iid $B_i \mid \theta \sim \mathcal{N}(\theta, 1)$; $\bar{B}$: mean. $C(B) = \bar{B} \pm 1.96/\sqrt{3}$. Interpret 95% confidence.

A. This realized interval has 95% posterior mass.

B. 95% of future individual readings lie inside it.

C. Repeated intervals cover fixed θ 95% of the time.

Confidence interval; coverage probability — [3] NIST §7.1.4 · Course example

EXERCISE

Sensor Calibration

Calibration: what does credibility describe?

Observed $b = (1, 2, 3)^T$; $\theta \mid b \sim \mathcal{N}(1.5, .25)$. A 95% credible interval describes which probability?

A. 95% posterior mass for θ, with b fixed.

B. 95% coverage for every fixed true offset.

C. 95% probability for a future noisy reading.

Posterior credible interval — [4] MIT 18.05 §2 · Course example

EXERCISE

Sensor Calibration

Calibration: why do the intervals differ?

Prior $\theta \sim \mathcal{N}(0, 1)$. Confidence $C(b) = 2 \pm 1.96/\sqrt{3}$; credible $I(b) = 1.5 \pm 1.96(.5)$. Why different?

A. Narrower credible interval means better coverage.

B. The zero-centered prior adds information.

C. The physical sensor became less noisy.

Confidence interval; credible interval — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6.1 · Course comparison

EXERCISE

Sensor Calibration

Calibration: new data, new prior

Readings $b = (2, 3)^T$ estimate offset θ; iid noise $\mathcal{N}(0, 1)$; prior $\theta \sim \mathcal{N}(4, 1)$. Which is valid?

A. The prior changes the data-only MLE.

B. The prior center can be ignored in MAP.

C. MAP moves above the data mean; MLE stays fixed.

MLE; MAP; confidence and credible intervals — [1] DL §§5.5–5.6 · [3] NIST §7.1.4 · [4] MIT 18.05 §2 · Course question

EXERCISE

Sensor Calibration

What does the sensor promise?

A specification says “±2 cm, 95%.” Which conclusion is justified without another definition?

A. It is a confidence interval for the offset.

B. It is a credible interval for the offset.

C. None: ask what quantity and probability model it describes.

Confidence; credibility; predictive uncertainty — [3] NIST §7.1.4 · [4] MIT 18.05 §2 · [1] DL §5.6, Eq.5.68 · Course question

EXERCISE

Synthesis

Which modeling bridge is valid?

Zero-mean iid Gaussian noise; isotropic Gaussian prior. Which chain gives the negative-log posterior?

A. Independence gives squares; Gaussian noise gives sums.

B. Gaussian noise gives squares; $-\log$ turns independent factors into sums.

C. A Gaussian prior removes the data residual term.

Gaussian likelihood; independence; MAP regularization — [1] DL §§5.5–5.6 · [1] DL §3.7 · Course question

References

Sources and optional extensions

[1] Deep Learning, chs.2–5 (2016).

[2] UBC CS340: Gaussian Distributions (2011).

[3] NIST/SEMATECH Statistical Handbook.

[4] MIT18.05: Probability Intervals (2022).

[5] Welch & Bishop: Kalman Filter (2004).

Definitions: [6] Stanford EE278 · [7] Data140 · [8] CMU Bayes notes · [9] Wisconsin Math225 · [10] MC Probability